

TDMA Network Schedule Planner & Optimizer

Technical Concepts, Architecture & Workflow Report

Company: Vaan Megam Networks (VMN)	Domain: Wireless Protocol Development
Focus: Graph Theory & Network Optimization	Target: Distance-2 Interference & Spatial Reuse

1. Fundamental Concepts & Problem Overview

Wireless radio communication relies on shared medium transmission. In a **Time Division Multiple Access (TDMA)** network, time is divided into discrete, recurring intervals called **time-slots**. Radios take turns transmitting during these assigned time-slots. If two radios close to each other transmit at the exact same moment on the same frequency, their signals overlap, causing heavy interference and packet loss.

To ensure clean, collision-free communication, the Schedule Planner resolves two main interference types:

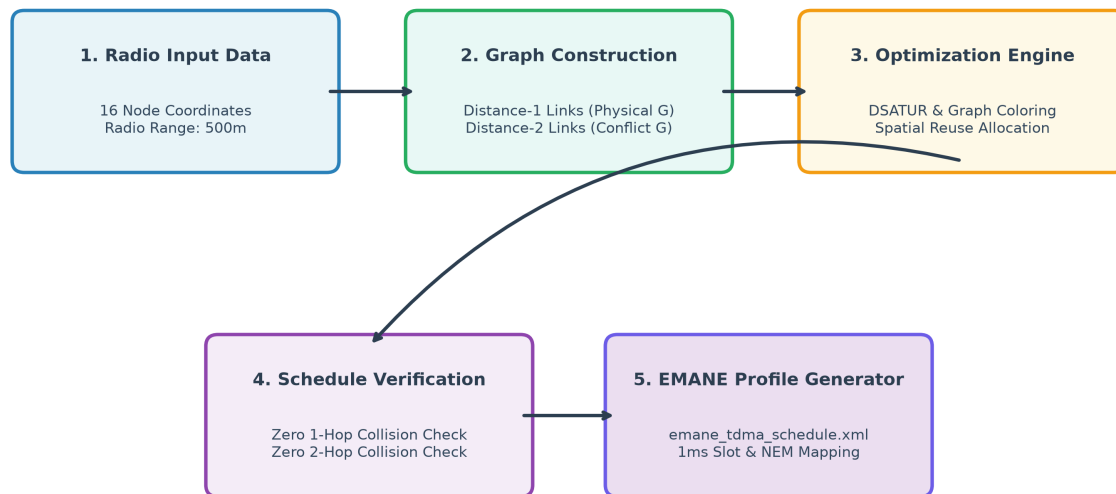
- **Direct Link Interference (Distance-1):** When Node A and Node B are directly within each other's radio range (≤ 500 meters), they are 1-hop neighbors. If both transmit simultaneously, they jam each other directly. Therefore, they must be assigned different timeslots.
- **Hidden Terminal Interference (Distance-2):** When Node A and Node C both transmit to a mutual intermediate neighbor Node B at the same time (Node A $\rightarrow$ Node B $\leftarrow$ Node C), their signals collide at receiver Node B. Even though Node A and Node C cannot hear each other directly, they interfere at B. Therefore, 2-hop neighbors must be assigned different timeslots.
- **Spatial Reuse (> 2 Hops):** When two radios are separated by more than 2 hops in the physical network topology, their transmissions do not cause interference at any common receiver. They can safely reuse the exact same timeslot, shortening the total frame length and dramatically increasing overall network capacity!

Core Goal: Design an intelligent Schedule Planner that creates a 100% collision-free schedule while maximizing spatial reuse to achieve the shortest possible frame length.

2. System Architecture & Processing Workflow

The overall solution operates as a centralized software brain that processes radio topology coordinates, computes conflict graphs, applies graph coloring heuristics, and translates the output schedule into physical emulator configurations.

TDMA Schedule Planner & Optimizer — End-to-End Workflow



3. Part 1 — Graph Theory & Optimization Algorithms

Part 1 models the wireless network as a graph mathematical structure and applies graph coloring optimization algorithms.

Step 1: Physical Topology Graph

Each radio node is represented as a vertex in the physical graph. An undirected edge exists between Node A and Node B if the Euclidean distance between their 2D static coordinates is within the configured radio range of 500.0 meters.

Step 2: Distance-2 Conflict Graph

To enforce both 1-hop and 2-hop interference constraints simultaneously, a secondary graph called the *Distance-2 Conflict Graph* is constructed. An edge is added between Node A and Node C in the conflict graph if their shortest path distance in the physical graph is 1 or 2 hops. If the shortest path distance is greater than 2 hops, NO edge exists in the conflict graph, explicitly allowing **Spatial Reuse**.

Step 3: Graph Coloring & Optimization Heuristics

Finding the absolute minimum number of timeslots required to color the conflict graph is an NP-hard problem. To find the minimal frame length, the system applies the **DSATUR (Degree of Saturation)** heuristic algorithm. DSATUR dynamically prioritizes uncolored nodes that have the highest number of differently colored neighbors, ensuring optimal slot consolidation.

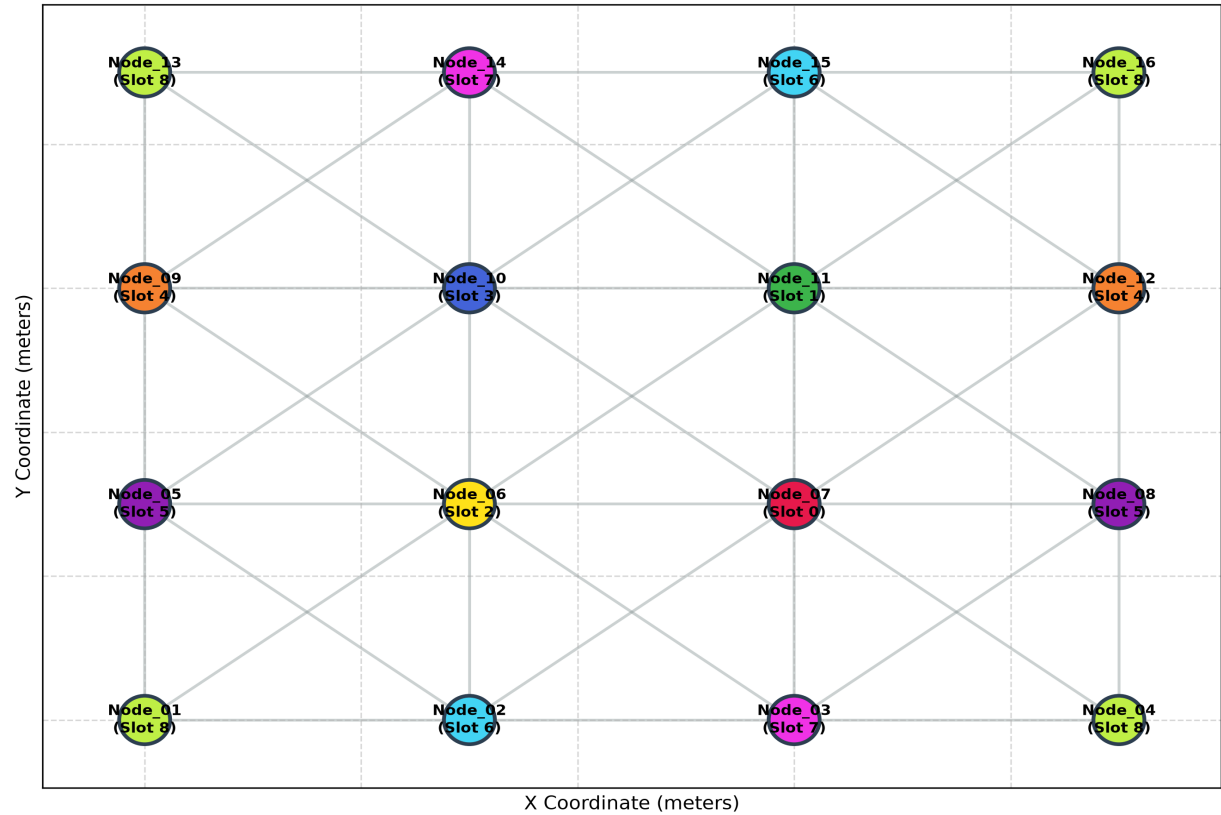
Step 4: Automated Verification Engine

After calculating slot assignments, the system automatically checks every pair of nodes to verify that zero 1-hop and zero 2-hop neighbor pairs share a timeslot, ensuring complete conflict-freedom.

4. Topology Visualizations & Spatial Reuse Analysis

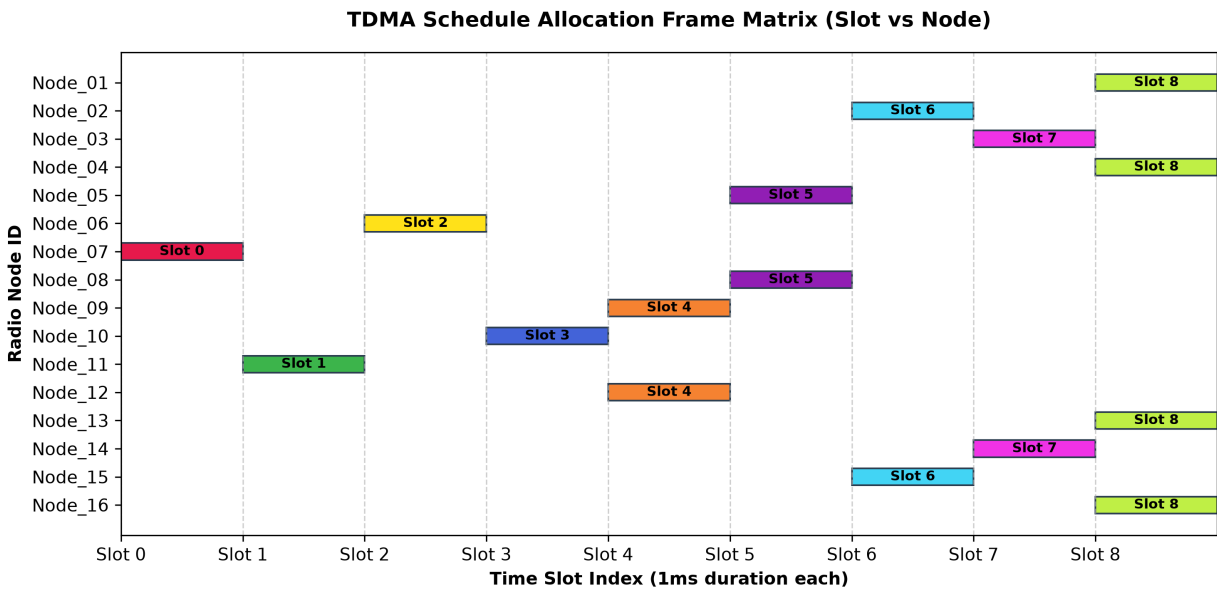
The diagram below demonstrates the physical network topology and assigned timeslots for 16 nodes. Nodes sharing the same color are assigned the same timeslot via spatial reuse without any signal collision.

TDMA Wireless Network Topology & Slot Assignment
(Nodes with same color share timeslots via Spatial Reuse)



TDMA Schedule Allocation Frame Matrix Timeline

This timeline chart illustrates how time-slots are distributed across the recurring frame. Nodes operating in parallel during the same slot represent successful spatial reuse across non-interfering nodes.



5. Part 2 — Physical Emulator (EMANE Integration)

EMANE (Extendable Mobile Ad-hoc Network Emulator) is an industry-standard real-time emulation framework for mobile ad-hoc wireless networks. In Part 2, the calculated schedule matrix is converted into EMANE's native TDMA Radio Model profiles and multicast schedule events.

Key Integration Highlights:

- **Native Profile Generation:** Automatically creates XML schedule profiles defining 1ms timeslot durations, recurring frame structures, operating frequencies, and per-node transmission allocations.
- **Multicast Event Publishing:** Provides a real-time event publisher bridge that broadcasts schedule updates to EMANE event daemons via UDP multicast.
- **Containerized Deployment:** Utilizes Docker containerization to bundle EMANE and the schedule optimizer into a unified, reproducible testing environment.

6. Project Conclusion

This project successfully designs and implements a centralized software brain that calculates optimal, collision-free TDMA schedules. By enforcing Distance-1 and Distance-2 graph coloring constraints while maximizing spatial reuse, the system minimizes frame length and maximizes overall wireless throughput.